

Performance Testing

Date	15 March 2026 1 July 2026
Team ID	XXXXXX 6a3a555ae5895b2c9759029d
Project Name	XXXXXX EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	[e.g., JMeter, Locust, k6, Postman]
Type of Testing	[e.g., Load Testing, Stress Testing, Spike Testing]
Target Module / API	[Specify which feature or endpoint was tested]
Test Environment	[e.g., Local / Cloud / Hosted Server] 5.02 ms (demo mode) Pass Avg
Test Date	overhead) 16.60 ms Pass FastAPI routing + DB write is highly performant performant 125 req/sec Pass Successfully passed sequential high-frequency tests high- 0% Pass No request errors or validation failures logged logged

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users 4%	Duration (sec) Pass	Expected Outcome Local lightweight JSON logging keeps CPU usage low CPU usage low
1				
2				
3				
4				

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds			
2	Response Time (Max)	< 5 seconds			
3	Throughput (Req/sec)				
4	Error Rate	< 1%			
5	CPU Utilization	< 80%			
6	Memory Utilization	< 80%			

Step 4: Observations & Analysis

Key Findings:					
[Describe what the			5.02 ms (demo mode)	Pass	Avg response is extremely fast (< 10 ms overhead) overhead)
			16.60 ms	Pass	FastAPI routing + DB write is highly performant
Bottlenecks Identified			125 req/sec	Pass	Successfully passed sequential high-frequency tests high-frequency tests
			0%	Pass	No request errors or validation failures logged
Optimization Steps			4%	Pass	Local lightweight JSON logging keeps CPU usage low CPU usage low

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]